



DPUSNDXWavPlayer

Plays WAV-files and optionally positions them in the sound simulation.

DPUSNDXWavPlayer uses OpenAL for sound playback and works fine with other SNDXDPU's, for example *DPUSNDXVstSource* and *DPUTRFSNDX*.

1. Inputs:

<i>Enable</i>	Samples will only be played if this input is not 0. default: 1
<i>Volume</i>	Volume of playback Range is 0 (silence) .. 1 (maximum). default: 1
<i>PlaySample[20]</i>	Corresponding sample is played if the input is not 0. Only one input of <i>PlaySample[20]</i> at a time should be set. default: 0
<i>Play</i>	Alternative playback: Index number of the sample to be played (1 to 20). default: 0
<i>Loop</i>	If set to non-zero value samples will be repeated endlessly default: 0

2. Inputs for 3D positioning:

<i>PositionMode</i>	Mode of the positioning. 0 (default): no positioning, standard stereo playback 1: absolute positioning in the SILAB world 2: relative positioning to listener
<i>X, Y, Z</i>	Position of the sound source. Either absolute (Position-Mode=1) or relative (PositionMode=2) to listener.
<i>vX, vY, vZ</i>	Velocity of the sound source. Absolute to the SILAB world coordinate axes (PositionMode=1) or relative to the listener (PositionMode=2)

3. Parameters:

<i>SoundPath</i>	Directory in which the samples are located. The path must be ended with a backslash „\“ (see functional principle). If no absolute path is set, the directory is searched within the SILAB install path default: „\silab\lib\sounds\wavplayer\“
<i>Sound[20]</i>	filename of the sample (see functional principle).



4. Functional principle:

This DPU is for WAV-files playback. A path is set in the configuration section of the DPU instantiation where the samples should be located. A set of files can be loaded by specifying their name in the configuration.

5. Example:

```
DPUSNDXWavPlayer WAVPlayer
{
    Computer = { SNDX };
    Index = 10;

    SoundPath = "e:\mysounds\";
    Sound01 = "horn.wav";
    Sound02 = "tireslip.wav";
    Sound03 = "crash.wav";
};
```

The samples are loaded into RAM before the start of the simulation. During the simulation playback can be done in two ways: either set Play to the index number of the sample or set one of the inputs of PlaySample[01...20] to 1.

Using more than one instance and playback of this DPU will cause the sounds to be mixed. Depending on the computers hardware there might be side effects like high latency.

6. Positioning:

Optionally the sound source can be positioned in the 3D environment.

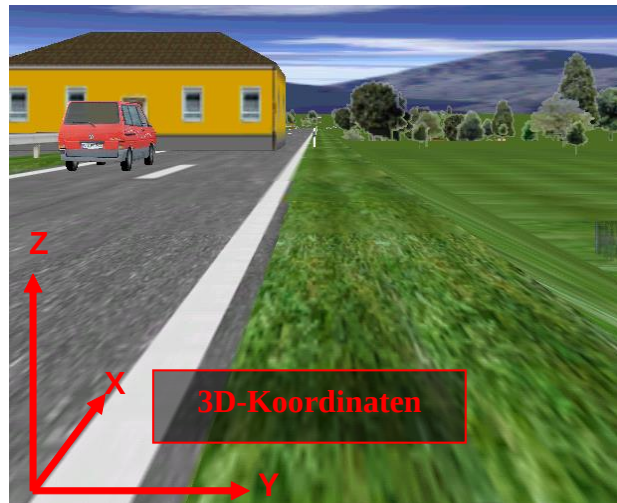
Note: Only mono channel WAV-files will be positioned correctly. Stereo or multichannel files force PositionMode=0.

Coordinates of the sound source are specified in the SILAB world coordinate system. The origin of this coordinate system is the setup point of the car during startup. At this time positive X-values point to the front, positive Y-values to the right and positive Z-values up. The vectors unit is meters. Positive angles (rad) along the (positive) axis are counter-clockwise.



SILAB DRIVING SIMULATION

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Using **PositionMode = 1** absolute coordinates for the sound source from this origin are expected. The velocity along the axes is needed to calculate any Doppler effect and is expected in meters per second.

Using **PositionMode = 2** relative coordinates for the sound source to the position of the listener (usually the car) are expected, as is the velocity. This mode is preferable for sound sources located in the car, i.e. driving with the test subject.

If no velocity is given for the inputs vX , vY und vZ) the velocity is calculated by numerical differential of the position (X , Y , Z) automatically. You have to make sure the position values are smooth enough (i.e. no sudden changes in values) and *DPUSNDXWavPlayer* gets enough calculation time (*Frequency* ≥ 50 Hz).